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GUIDELINES - CONSENSUS-BASED

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Prevention and treatment of venous thromboembolism in patients with multiple myeloma: Clinical practice guidelines on behalf of the European Myeloma Network

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Abstract

Venous thromboembolism (VTE) is a frequent complication in patients with multiple myeloma (MM) that leads to increased morbidity, mortality, treatment interruptions, and reduced quality of life. An Expert Panel Consensus Guideline on behalf of the European Myeloma Network provides evidence-based recommendations for VTE prevention and treatment in patients with MM. Key recommendations include the following:

- *Patient education*. Informing patients about VTE risk and signs/symptoms of deep vein thrombosis and pulmonary embolism.
- *Specialized care*. Establishing a dedicated diagnostic and therapeutic pathway for the prevention and treatment of VTE that occurs in an outpatient setting.
- *Risk assessment*. Regular evaluation of VTE risk using validated scores.
- *Primary thromboprophylaxis*. Initiating pharmacological thromboprophylaxis with low-molecular-weight heparin (LMWH) or specific direct factor Xa inhibitors (anti-Xa agents) in high-risk patients.
- *Prompt diagnosis and treatment*. Early diagnosis of VTE and immediate initiation of antithrombotic therapy with LMWH or anti-Xa agents.
- *Secondary prophylaxis*. Personalized long-term antithrombotic therapy based on individual profiles of risk of VTE recurrence or bleeding.

By implementing these guidelines, healthcare providers can improve the prevention and treatment of VTE in patients with MM, leading to better clinical outcomes and quality of life.

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INTRODUCTION

Advances in the treatment of multiple myeloma (MM), including patients with newly diagnosed or relapsed/refractory disease, have significantly improved mortality and quality of life. The median survival of patients with MM has doubled from 3–4 to 8–10 years in the era of novel therapeutic strategies.[1]

Venous thromboembolism (VTE), including deep vein thrombosis (DVT), pulmonary embolism (PE), and atypical thrombosis (e.g., cerebral or splanchnic veins), is a common and serious complication in MM. Its occurrence exacerbates morbidity and undermines therapeutic gains by complicating treatment regimens, delaying therapy, and increasing mortality risk. The risk of VTE is driven by a complex interplay of MM-specific risk factors (such as hypercoagulability, immunomodulatory drug [IMiD] use, and disease-related endothelial dysfunction) and patient-related risk factors.

The incidence of VTE in patients with MM ranges between 8 and 22 per 1000 person-years.[2] A competing risk analysis of 13,700 MM patients in the United States revealed a VTE incidence of 1050 events over a 1-year follow-up period.[2] VTE rates reported in clinical trials range from 4% to 16%.[3,4] Overall, patients with MM have a ninefold increased risk of VTE compared to the general population. About 50% of patients present with symptomatic distal DVT or central venous catheter (CVC)-related thrombosis (CRT), whereas 15% suffer a PE. Superficial vein thrombosis and mesenteric or upper limb vein thrombosis have also been reported. In newly diagnosed MM (NDMM) patients, more than half of the VTE events occur within the first 3–6 months from initial diagnosis and treatment initiation. Similarly, the risk of VTE significantly increases within the first 3–6 months after documented treatment relapse or progression.[5,6]

Monoclonal gammopathy of undetermined significance (MGUS), a precursor/premalignant condition, has also been shown to increase the risk of VTE.[7] In a recent multicenter cohort study of 1491 patients with MGUS, the incidence of VTE was 1.9 per 1000 person-years (2.7% during follow-up).[7] The Iceland Screens, Treats, or Prevents MM (iSTOPMM) study confirmed that MGUS is associated with a 1.3-fold increased risk of VTE (hazard ratio [HR] 1.31; 95% confidence interval [CI]: 1.08–1.57) compared to healthy controls. After a median follow-up of 3.7 years, the rate of VTE was 10.3 per 1000 person-years in the MGUS group versus 6.0 per 1000 person-years in the non-MGUS group.[8]

Large database studies have demonstrated a two- to threefold higher risk of mortality in patients with MM and VTE.[9,10] A recent retrospective study in 672 MM patients who received lenalidomide-based triplets or quadruplets suggested a survival disadvantage for patients who experienced VTE (HR = 2.61, 95% CI = 1.74–3.9). However, disease burden and other risk factors related with MM confounded this finding.[4] Achieving effective VTE prevention and management would improve quality of life, reduce VTE-related morbidity, and healthcare expenditures.[11,12]

Aiming to contribute to optimal care for patients with MM, this *Expert Panel Consensus on Practice Guidelines for the Prevention and Treatment of Venous Thromboembolism in Patients with Multiple Myeloma* provides evidence-based recommendations. These are tailored to the needs of hematologists, oncologists, internists, angiologists, general practitioners, allied health professionals, and policymakers who are involved in the care of patients with MM. By implementing these guidelines, physicians can effectively prevent and treat VTE in MM patients and reduce the overall burden of VTE.

METHODOLOGY

An interdisciplinary panel of experts on cancer-associated thrombosis (CAT) and experts on MM from different institutions and countries in Europe, on behalf of the European Myeloma Network (EMN), developed these recommendations. The guideline development and structure followed the Population, Intervention, Control, and Outcomes (PICO) format. We electronically searched major databases (i.e., Cochrane Central and MEDLINE, PubMed) to identify relevant systematic reviews, randomized controlled trials (RCTs), observational studies, and case series. These electronic searches were performed looking for studies published in English from inception to December 2024. For each PICO question, all potentially eligible studies were assessed for eligibility according to prespecified criteria. Content experts were asked to indicate any additional studies not identified by the search (see Supplementary Material 1).

After the manuscript was prepared, it was discussed at the EMN Trialist Annual Meeting (Baveno, Italy, November 5–6, 2023 and 2024), followed by two rounds of revisions among the panel members.

Recommendation formulation. We used the principles outlined in the evidence to decision (EtD) framework to formulate recommendations.[13] The EtD framework addresses several key domains, including priority setting, the magnitude of benefits and harms, certainty of the evidence,

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patient values, the balance of desirable and undesirable effects, resource use and costs, equity, acceptability, and feasibility (see Supplementary Material 2). The wording "we recommend" is used for strong recommendations (recommendation Grade 1A or 1B) and "we suggest" for weak recommendations (recommendation Grades 2A–2C). The implications of the recommendation strength are presented in Table 1.

Primary thromboprophylaxis in patients with MM

Risk factors associated with VTE

The identification of patients at high risk for VTE is critical for the effective and safe use of pharmacological thromboprophylaxis in patients with MM. Patients should be assessed for VTE risk at diagnosis, at relapse, and systematically during treatment. Patients must be stratified as high-risk and intermediate/low-risk for VTE, enabling targeted prevention strategies that balance the benefits of pharmacological thromboprophylaxis with the potential bleeding risk.

Predictors of VTE have been identified from population-based databases, registries, hospital records, retrospective cohorts, prospective observational studies, and clinical trials.

The global risk is influenced by the following:

i. *Disease-related risk factors*. MM is associated with hypercoagulability and endothelial cell activation. The time since diagnosis (NDMM patients are at higher risk of VTE compared to patients at relapse, as tumor burden is often highest at initial diagnosis), the tumor burden, immobilization (due to bone disease or any other reason), levels, and type of paraproteinemia are predictors of MM-associated thrombosis. Monoclonal paraproteinemia may cause hyperviscosity, hyperfibrinogenemia, decreased activity of the natural coagulation inhibitors protein C or protein S, acquired resistance to activated protein C, or impaired fibrinolysis. Cytokines such as interleukin-6 and tumor necrosis factor (TNF), as well as plasma cell-derived extracellular vesicles, can cause endothelial cell activation and platelet activation; all of which are associated with the risk of VTE.[14–16]
ii. *Antimyeloma therapy*. IMiDs (thalidomide and its derivatives lenalidomide and pomalidomide, likewise iberdomide and mezigdomide to come in the near future) increase the risk of VTE. The rate of VTE in patients on thalidomide or lenalidomide monotherapy is 3%–4%. The addition of high-dose dexamethasone, or multiagent chemotherapy, increases VTE risk up to 26%.[17,18] Although reducing the dose of dexamethasone lowers the VTE risk, it still remains substantial at around 10%.[19] In addition, the risk of VTE remains significantly elevated, independently of the duration of treatment exposure.[20] Pomalidomide is associated with a lower risk of VTE compared to lenalidomide, with reported VTE rates being dose-dependent and ranging between 3% and 7% in patients receiving a 4-mg dose.[21] Treatment with the proteasome inhibitor bortezomib is associated with low VTE risk.[22] Carfilzomib induces hypercoagulability and is associated with an increased VTE risk.[23] In a meta-analysis involving 924 patients treated with carfilzomib/lenalidomide/dexamethasone (KRd), the VTE rate was 8.9% versus 6.2% among 1380 patients treated with bortezomib/lenalidomide/dexamethasone (VRd).[24] The study reviewed one randomized trial and five retrospective studies; VTE risk assessment using prespecified risk assessment tools was performed

TABLE 1 Grades of recommendations and implications in clinical practice.

Recommendation grade	Meaning	Implications to patients	Implications to clinicians	Implications to policymakers
1. Evidence strongly suggests that the benefit of the procedure outweighs the potential risks or risks of the procedure outweigh the potential benefits	A: Consistent evidence from systematic reviews or high-quality randomized studies or from high-quality observational studies	Must do or must avoid	Almost all individuals in this situation would want the recommended intervention, and only a small proportion would not want it	Most individuals should receive the recommended course of action
				Can be adapted as policy in most situations, including the use as performance indicators
2. Evidence suggests the benefit and risk of the procedure are finely balanced and uncertain	B: Evidence from randomized and observational studies with important methodological flaws C: Evidence from randomized and observational studies with major methodological flaws and other sources of evidence (i.e., case series)	Consider doing or consider avoiding	The majority of individuals in this situation would want the recommended intervention, but many would not	Different choices are likely to be appropriate for different patients, and recommendation should be tailored to individual patients' circumstances (i.e., patients', family's, or substitute decision-maker's values and preferences)
				Policies will likely be variable

Note: This table outlines the grading system used to assess the strength and certainty of recommendations in the guideline. Grade 1 recommendations, supported by strong evidence (A), are associated with broad clinical consensus and expected to be adopted widely by patients, clinicians, and policymakers. Grade 2 recommendations (B/C) reflect greater uncertainty, requiring individualized decision-making based on patient preferences and clinical judgment.

only in two studies, Piedra et al.[3] used the SAVED-VTE risk score, and Loncharich et al.,[25] the IMPEDE score to stratify patients. In a retrospective study[4] ($n = 672$), the incidence of VTE was 10.6% in patients treated with KRd, 8.9% in patients who received daratumumab-lenalidomide-dexamethasone plus a proteasome inhibitor, and 7% in patients treated with VRd. In the study by Piedra et al. ($n = 305$), VTE incidence was 16.1% in patients who received KRD plus aspirin (ASA), 4.8% in patients who received KRD plus rivaroxaban, and 4.8% in patients who received VRD plus ASA.[3] The CARDAMON trial reported thrombotic microangiopathy in 2.8% of patients who received carfilzomib-cyclophosphamide-dexamethasone.[26] In the EMN09 study, the non-IMiD containing regimen of carfilzomib-bendamustine-dexamethasone combination resulted in a VTE rate of 10%.[27] The anti-CD38 monoclonal antibody daratumumab and other monoclonal antibodies, such as belantamab mafodotin or elotuzumab, and the oral proteasome inhibitor ixazomib have not been associated with an amplification of the thrombotic risk.[28] In the GRIFFIN study, daratumumab with VRD (D-VRd) was compared to VRd, and the all-grade cumulative rates of VTE were 10% versus 15%, respectively.[29] As per protocol, antithrombotic prophylaxis was administered in accordance with IMWG guidelines, with 80% of patients receiving acetylsalicylic acid, 15% receiving LMWH, and 5%–8% receiving direct factor Xa inhibitors. A post hoc pooled analysis of the Phase III CASTOR, POLLUX, and MAIA trials demonstrated that daratumumab treatment did not increase VTE risks.[30] Administration of bispecific antibodies is not associated with an increase in the VTE risk. A meta-analysis of 47 studies enrolling 7040 patients with hematologic malignancies, including MM, who received CAR T-cell therapy, showed that the overall rate of VTE during the first 6 months after CAR T-cell administration was 2.4%, whereas the rate of any bleeding event was 1.9%. The incidence of arterial thrombosis ranged from 0.3% to 0.8%.[31]

iii. *Patient-related intrinsic risk factors*. Advanced age (particularly in patients older than 70 years), female gender, personal history of VTE, presence of known hereditary thrombophilia (factor V Leiden or factor II G20210A mutations, antithrombin, protein C or protein S deficiency) or other gene polymorphisms (i.e., ABO locus), cardiovascular risk factors (diabetes, obesity, hypertension, and dyslipidemias) or cardiovascular disease (ischemic stroke, coronary artery or carotid artery disease, and peripheral artery disease), inflammatory bowel disease, autoimmune diseases, and renal insufficiency, figure among the most common and the most important risk factors for CAT.[32]

iv. *Triggering risk factors*. Central vein catheters, use of erythropoietin or other colony-stimulating factors, recent hospitalization (less than 3 months) or hospitalization for acute medical illness, immunosuppression, surgical interventions, and immobilization (for any reason) are among the major triggering risk factors for VTE in patients with MM.[33]

Risk assessment models for VTE in patients with MM

Individualized VTE risk evaluation to identify patients eligible for pharmacological thromboprophylaxis is a clinical need as recommended by the recent ESMO *Clinical Practice Guideline for the Prevention and Treatment of VTE in Cancer Patients*.[34] The guidelines published by the International Myeloma Working Group (IMWG),[35] the EMN,[36] and the National Comprehensive Cancer Network (NCCN)[37] adopted a simplified algorithm, in practice since 2014, based on the concept that IMiD-based therapy is the primary determinant of VTE risk. Nevertheless, this algorithm does not stem from a derivation cohort analysis but is based on an empirical scoring of the VTE risk predictors. According to the IWMG algorithm, patients are categorized into two risk groups: low-risk (0–1 risk factor) and high-risk (≥2 risk factors) (Table 2). The IWMG risk stratification is not accurate, leading to an underestimation of the VTE risk. A high incidence of VTE in the "low risk" group of patients has been systematically documented by clinical studies and case series.[38]

To improve VTE risk stratification in MM, three validated scoring systems have emerged: IMPEDE-VTE, SAVED, and PRISM (Tables 2 and 3). These tools represent significant advances in risk prediction, though each has distinct clinical applications and limitations, which are presented herein.

The IMPEDE-VTE score was derived from analysis of the retrospective database of the Veterans Administration Cancer Registry including 4446 patients and was validated at the SEER-Medicare database.[39] The IMPEDE-VTE score stratifies patients into low-, intermediate-, and high-risk groups. The reported rates of VTE at 6-month follow-up were 3.3% in the low-risk, 8.3% in the intermediate-risk, and 15.2% in the high-risk group. The IMPEDE-VTE score showed improved predictive accuracy (c-statistic = 0.64), as compared to the IMWG/NCCN model (c-statistic = 0.55). The IMPEDE-VTE score was derived from an unselected, real-world MM population, making it broadly applicable across diverse clinical scenarios—including newly diagnosed chemotherapy-naïve patients, those undergoing active myeloma treatment, and patients with disease progression or relapse. This generalizability stems from its development in heterogeneous cohorts without restrictive enrollment criteria.

The SAVED score was derived from a retrospective database analysis of 2397 patients treated with IMiDs available at the SEER-Medicare database (Tables 2 and 3).[40] The SAVED score's applicability is inherently limited to patients receiving IMiDs, as its derivation cohort exclusively comprised IMiD-treated patients. The SAVED score stratifies patients at low-risk and high-risk for VTE. The predictive accuracy of the SAVED score was slightly higher (c-statistic = 0.6) as compared to the IMWG/NCCN model (c-statistic = 0.55).

The PRISM score was derived from a retrospective database of 934 patients treated at the Cleveland Clinic.[41] It stratifies patients into low-, intermediate-, and high-risk groups (Tables 2 and 3). The 12-month VTE rates were 2.7%, 10.8%, and 36.5%, respectively. The external validation of the PRISM score in a group of 257 patients at Columbia University Irving Medical Center showed a low predicted accuracy (c-statistic of 0.59), though slightly improved as compared to the IMWG/NCCN algorithm (c-statistic = 0.55).

The IMPEDE-VTE, SAVED, and PRISM scores, as compared to the IMWG/NCCN algorithm, represent important advances in identifying MM patients at risk for VTE since they have been derived from well-defined cohorts and have been validated.

In summary, the comparison of the three scores (Table 2) highlights several important points:

i. The IMPEDE-VTE and PRISM scores can be applied to any patient with MM, whereas the SAVED score is specifically designed for patients receiving treatment with IMiDs.
ii. The applicability of the PRISM score is limited by its reliance on the assessment of metaphase cytogenetics, which are not routinely performed. Moreover, the generalizability of the PRISM score is questioned since the score is derived from a single-center cohort study.

TABLE 2 Risk assessment models available for the evaluation of VTE in patients with MM.

	IMPEDE-VTE	SAVED score[a]	PRISM	IMWG
Population	*N* = 4446	*N* = 2397	*N* = 783	
VTE history	+5	+3	+8	+1
Dexamethasone				
High dose (>160 mg/cycle)	+4	+2		+2 for >480 mg/cycle plus IMiD
Low dose (<160 mg/cycle)	+2	+1		+ 1 for <480 mg/cycle plus IMiD
IMiDs	+4		+2	+1
Fracture pelvis/hip/femur	+4			+1
Use of recombinant epoetin	+1			+1
Tunneled line/CVC	+2			+1
BMI	+1 For ≥25 kg/m^2			+1 for BMI > 30 kg/m^2
Race	−3 Asian/Pacific Islander	−3 Asian race	+1 Black race	+1 Caucasian
Age ≥ 80 Age > 75		+1		+1
Surgery < 90 days		+2	+5	+1
Existing use of anticoagulant or antiplatelet therapy	−4 (−3 if prophylactic)			
Abnormal metaphase cytogenetics			+2	
Other therapy	+3 doxorubicin			+2 MiD plus multiagent chemo OR doxorubicin
New MM diagnosis evidence of hyperviscosity				+1 +1
Other comorbidity, hospitalization, PS > 1, thrombophilia				+1
Risk stratification	Score ≤ 3: low-risk Score 4–7: intermediate-risk Score ≥ 8: high-risk	Score ≤ 1: low-risk Score ≥ 2: high-risk	Score 0: low-risk Score 1–6: intermediate-risk Score ≥ 7 high-risk	Score 0–1: low-risk Score > 1: high-risk

Note: This table shows the four VTE risk assessment tools (IMPEDE-VTE, SAVED, PRISM, and IMWG) used in patients with multiple myeloma, highlighting the variables included and how patients are stratified into risk categories.

Abbreviations: BMI, body mass index; CVC, central venous catheter; IMiDs, immunomodulatory agents; IMPEDE-VTE, immunomodulatory agent, medication history, pelvic, hip or femur fracture, erythropoetin stimulating agent, dexamethasone/doxorubicin, ethnicity/race, venous thromboembolism history, tunelled line/central venous catheter, existing thromboprophylaxis; IMWG, International Myeloma Working Group; MM, multiple myeloma; VTE, venous thromboembolism; PRISM, prior VTE, race (black), IMiD use, surgery within 90 days, metaphase cytogenetics; PS, performance status; SAVED, surgery, age, venous history, extremity edema, dexamethasone.

[a]Developed only in patients who received IMiDs.

Common limitations across all models and perspectives. All current RAMs are derived from retrospective cohorts that primarily included patients treated with older therapeutic regimens, predominantly doublets, rather than the triplet or quadruplet combinations that are now standard of care in patients with MM. Additionally, the derivation cohorts for the IMPEDE-VTE and SAVED scores mostly involved patients receiving high-dose dexamethasone (≥160 mg/cycle), which is no longer considered standard of care. Newer agents such as carfilzomib and daratumumab are underrepresented or entirely absent from these datasets.[42]

Given these limitations, there is a need to update or recalibrate the RAMs by evaluating them in cohorts of patients who receive newer antimyeloma therapies. There is also a need for improving the predictive accuracy of the scores. Lastly, the positive predictive value of the current clinical scores remains modest—approximately 15%—highlighting the need for more precise stratification tools. Incorporating specific biomarkers of hypercoagulability offers a promising strategy to enhance the accuracy of the score and particularly the positive predictive value. Ongoing prospective studies are evaluating the clinical utility of biomarkers such as D-dimer, thrombin generation, procoagulant phospholipid-dependent clotting time, P-selectin levels, and circulating procoagulant microvesicles, as well as markers of endothelial cell activation. Integrating these clinically validated biomarkers with established clinical RAMs is expected to improve both the accuracy and reliability of VTE risk assessment in patients with MM.

Taking into consideration the ensemble of the above points, and acknowledging the limitations, among the so far available scores for VTE risk assessment, the IMPEDE-VTE score fulfills the minimum criteria for assessment of all patients with MM, given its comprehensive inclusion of relevant risk factors, applicability across treatment regimens, reliance on major available clinical predictors and better accuracy compared either to the IMWG/NCCN or to the SAVED or the PRISM score.

Thromboprophylaxis strategies

Clinical trials designed to evaluate the efficacy and safety of primary pharmacological thromboprophylaxis in patients with MM are limited.

Current IMWG guidelines recommend low-dose ASA for thromboprophylaxis in low-risk MM patients. Cumulative evidence

TABLE 3 Comparison of VTE risk scores in multiple myeloma.

Characteristic	IMPEDE-VTE	SAVED	PRISM	IMWG
Population used for development	*N* = 4446 MM patients	*N* = 2397 MM patients on IMiDs	*N* = 783 newly diagnosed MM patients	Expert opinion consensus
Database/cohort	Veterans Health Administration (VHA) database, United States	SEER-Medicare database, United States	Canadian Myeloma Research Group (CMRG) database	N/A (consensus guidelines)
Study design	Retrospective	Retrospective	Retrospective	Not applicable
C-index (discrimination)	0.65 (derivation), 0.63 (validation)	0.61	0.67	Not applicable
External validation	Yes (external cohort of 1747 patients)	Yes (validated in 1181 patients)	Yes	Not applicable
Sensitivity/specificity	Sensitivity: ~71%, specificity: ~52%	Sensitivity: ~72%, specificity: ~48%	Sensitivity: 77%, specificity: 57%	Not applicable
Variables included	IMiD use, BMI, fracture, epoetin use, tunneled line, Asian race, age, recent surgery, thromboprophylaxis, cytogenetics, doxorubicin	IMiD use, high-dose dexamethasone, prior VTE, Asian race, BMI ≥ 25	Fracture, BMI > 30, race (Black/Caucasian/Asian), cytogenetics, surgery, doxorubicin, IMiD + high-dose steroid, hyperviscosity, PS > 1, comorbidities	IMiD/steroid use, ≥2 individual or treatment-related risk factors (e.g., VTE history, CVC, surgery, comorbidities)
Risk categories	Low (≤3), intermediate (4–7), high (≥8)	Low (≤1), high (≥2)	Low (0), intermediate (1–6), high (≥7)	Low (0–1), high (>1)
Limitations	Derived retrospectively; no daratumumab/carf-based therapies; limited in modern regimens	Applies only to IMiD users; developed before anti-CD38 therapies; modest PPV	Includes metaphase cytogenetics (not widely available); moderate PPV; limited generalizability	Subjective; not validated; lacks granularity and risk score weighting
Validation prospectively in clinical trial	No	No	No	No

Abbreviations: BMI, body mass index; CVC, central venous catheter; IMiDs, immunomodulatory agents; IMPEDE-VTE: immunomodulatory agent, medication history, pelvic, hip or femur fracture, erythropoetin stimulating agent, dexamethasone/doxorubicin, ethnicity/race, venous thromboembolism history, tunelled line/central venous catheter, existing thromboprophylaxis; IMWG, International Myeloma Working Group; MM, multiple myeloma; PPV, positive predictive value; PRISM, prior VTE, IMiD use, surgery within 90 days, methaphase cytogenetics; SAVED, surgery, age, venous history, extremity edema, dexamethasone; VTE, venous thromboembolism.

from clinical studies demonstrates that ASA provides inferior protection against VTE compared to LMWH, with relative risk (RR) reductions of approximately 50% versus 70%–80%, respectively, in IMiD-treated populations.[43,44] However, given the established safety profile and potential benefits, low-dose aspirin remains a reasonable thromboprophylaxis option for MM patients classified as low-risk for VTE. A meta-analysis in 439 outpatients with MM showed that thromboprophylaxis with LMWH significantly reduced the risk of symptomatic VTE compared to vitamin K antagonists (VKAs) RR 0.33, 95% CI 0.14–0.83; high-certainty evidence). In 781 patients, LMWH significantly decreased the rate of symptomatic VTE compared to ASA (RR 0.51, 95% CI 0.22–1.17; 2 studies, moderate-certainty evidence).[40] Major bleeding was not observed in patients treated with LMWH or VKA and was reported in less than 1% of patients treated with ASA.

A more recent systematic review identified only four randomized clinical trials (RCTs) including in total 1042 patients that evaluated the efficacy and safety pharmacological thromboprophylaxis in patients with MM.[45] Two studies compared low-dose ASA to prophylactic LMWH.[46,47] One study evaluated VKA versus LMWH or ASA, and the fourth compared low-dose ASA versus direct anti-Xa agents.[3,48] The certainty of evidence comparing the efficacy and safety of ASA, VKA, and LMWH was low to very low. This uncertainty may stem from the limited accuracy of the risk assessment models (RAMs) used for patient stratification and assignment of thromboprophylactic regimens. These methodological shortcomings likely contributed to the high heterogeneity across studies and the inability to demonstrate a clear therapeutic superiority of one intervention over another.

A systematic literature review identified 10 RCTs and observational studies in patients with MM ($n = 1026$) who received primary thromboprophylaxis with direct anti-Xa agents or ASA. Thromboprophylaxis with direct anti-Xa agents was associated with a lower incidence of VTE as compared to that with ASA (odds ratio [OR] 0.33; 95% CI 0.16–0.68; $P < 0.001$), whereas the bleeding events did not differ significantly between the two groups.[49] The authors recommend caution due to the heterogeneity of population characteristics and potential bias of the observational studies included. An exploratory meta-analysis of studies in patients with MM investigated thromboprophylaxis with ASA versus no intervention or prophylactic dose of LWMH. Ten studies involving 1964 patients were analyzed.[50] In several longitudinal studies and two RCTs, ASA reduced the risk of VTE significantly compared to no intervention (OR = 0.20; 95% CI: 0.07–0.61; $P = 0.005$; $I^2 = 41\%$) but was associated with higher VTE risk compared to LMWH (OR = 2.60; 95% CI: 1.08–6.25; $P = 0.03$; $I^2 = 0\%$). An older prospective observational study (MELISSE trial), which enrolled patients with MM treated with IMiDs, reported a 7% VTE rate in patients who received ASA versus 3% in those treated with LMWH and none in patients treated with VKA.[38]

Thromboprophylaxis with direct anti-Xa agents in outpatients with MM is an attractive strategy due to the convenient mode of administration, the wide therapeutic window, and the limited drug-to-drug interactions compared to VKA. The AVERT and CASSINI trials randomized 574 and 841 intermediate/high-risk patients for CAT (based on a value of ≥2 on the Khorana risk score) to either placebo or apixaban 2.5 mg b.i.d. or rivaroxaban 10 mg o.d., respectively. In the AVERT trial, apixaban 2.5 mg b.i.d. for 180 days was associated with a lower VTE rate than placebo (4.2% vs. 10.2%; HR: 0.41 [95% CI 0.26–0.65; $P = 0.0001$]; number needed to treat = 17). The rate of major bleeding was higher in patients in the apixaban arm (3.5% vs. 1.8%; HR 2.00 [95% CI 1.01–3.95; $P = 0.046$]; number needed to harm = 59).[51] In the CASSINI trial, administration of rivaroxaban 10 mg o.d. for 180 days did not show a significant difference in VTE risk compared to placebo. Still, VTE rates were significantly lower with rivaroxaban in the on-treatment analysis (2.6% vs. 6.4% HR 0.40 [95% CI 0.2–0.8]). The rate of major bleeding was low and comparable in both groups.[52] Neither study included patients with major thrombocytopenia (platelet count $<50 \times 10^9$/L) or severe renal impairment (creatinine clearance [CrCl] < 30 mL/min). Of note, only a limited number of patients with MM were included in these studies. Three pilot studies[53-55] and a retrospective one, enrolled MM patients and showed promising results for the efficacy and safety of thromboprophylaxis with apixaban 2.5 mg b.i.d. for at least 6 months.[3] Among 305 patients, only two VTE events were reported (0.6%). A retrospective study of 305 patients with NDMM showed that thromboprophylaxis with rivaroxaban 10 mg o.d. can be an effective and safe means of thromboprophylaxis.[3] In patients treated with KRD plus rivaroxaban, the VTE rate was 4.8% compared to 16.1% in patients treated with ASA, similar to that of patients treated with VRD and ASA (4.8%). A small study of 37 NDMM patients or relapsed disease compared the efficacy and safety of thromboprophylaxis with warfarin at doses adapted to achieve an international normalized ratio (INR) of 2–3 or apixaban 2.5 mg twice daily for 3 months. No VTE events were observed in the warfarin group. One VTE episode was reported in the apixaban group, major hemorrhagic episodes in two and four patients, respectively.[56] Another study in 70 NDMM patients on IMiD-based regimens reported no cases of VTE among those receiving apixaban 2.5 mg twice daily. However, two arterial events and one major bleeding episode were noted, highlighting the need for ongoing evaluation of anticoagulation strategies in this patient population.[55] Overall, data point to equivalent efficacy and safety of direct anti-Xas compared to LWMH, which justifies the use of either in this setting.

Nevertheless, there is a need for Phase III RCTs evaluating the efficacy and safety of thromboprophylaxis with direct anti-Xa agents or LMWHs in high-risk patients with MM.

Duration of thromboprophylaxis

With improvement in survival rates for MM patients, a greater focus on the long-term management of VTE risk is needed. NDMM patients receive prolonged-duration lenalidomide maintenance therapy following induction, and patients at relapse continue treatment until disease progression or unacceptable toxicity. Regular evaluation of the benefit–risk ratio of extended pharmacological thromboprophylaxis is therefore required. Lenalidomide maintenance carries a continuous VTE risk between 4% and 6%.[57] A post hoc analysis of the Griffin trial showed that VTE risk is highest at induction but continues to increase over time (2.2 for daraVRD and 7% for VRD at maintenance between Cycles 7 and 18 and 3.6% and 1.7%, respectively, at maintenance 18+).[58] At subsequent treatment lines, even when the tumor burden decreases, the VTE risk persists.

Key steps in optimizing the benefit–risk ratio of extended pharmacological thromboprophylaxis in patients with MM include the following:

i. Implement validated RAM (e.g., IMPEDE-VTE) and regular reassessment of VTE risk.
ii. Consider treatment-specific risks (IMiDs, high-dose dexamethasone, etc.) versus.
iii. Regular reassessment of thrombotic/bleeding risks
   - evaluation of bleeding risk predictors: anemia, thrombocytopenia, renal and liver function, and drug-to-drug interactions.
iv. Regular evaluation of patients' preference, patients' adherence, and the impact of extended thromboprophylaxis on the quality-of-life indexes.
v. Patient education on

- early recognition of symptoms and signs of VTE,
- early recognition of symptoms and signs of hemorrhagic diathesis, and
- clear patients' pathway in case of bleeding or VTE.

Recommendations

1. *We recommend to provide in patients with M with educational materials on VTE. These materials should include comprehensive details on the burden of VTE and its consequences in patients with MM, on the risk factors of VTE, the clinical signs and symptoms of VTE, a guidance to enhance patient awareness as well as information for communication with the treating physicians specialized for the management of VTE and antithrombotic treatment* ***(Grade 1A)****.*
2. *We recommend the implementation of systematic VTE risk assessment for all MM patients at the* initial diagnosis, treatment initiation, relapse/progression, and whenever treatment regimens are modified ***(Grade 1A)****.*
3. *We recommend the implementation of the IMPEDE-VTE score for VTE risk assessment in all patients with MM; given its comprehensive inclusion of relevant risk factors, applicability across treatment regimens, and reliance on universally available clinical parameters. The IMPEDE-VTE score should be applied within the context of its limitations* ***(Grade 1B)****.*
4. *We recommend the stratification of patients into two risk groups as either high/intermediate or low risk for VTE* ***(Grade 2A)****.*
5. *Patients who receive IMiD-based regimens or carfilzomib combinations should be stratified as high/intermediate risk for VTE* ***(Grade 2B)****.*
6. *We suggest risk assessment for VTE upon primary diagnosis and antimyeloma treatment initiation, following induction completion or disease remission, at diagnosis of disease progression or recurrence, when modifications of therapeutic lines are scheduled* ***(Grade 2B)****.*
7. *We recommend the administration of pharmacological thromboprophylaxis with direct anti-Xa agents or LMWH to patients classified at high/intermediate risk for VTE* ***(Grade 1B)****.*
   a. *Antithrombotic agents suggested for thromboprophylaxis in high/intermediate-risk patients with MM are (see* Table 4 *for dosing guidelines):*
      i. *LMWH, o.d. subcutaneous injection in prophylactic doses approved by health authorities and as recommended by the manufacturer.*
      ii. *Rivaroxaban 10 mg o.d. p.o.*
      iii. *Apixaban 2.5 mg b.i.d. p.o.*
      iv. *We recommend to consider apixaban 2.5 mg b.i.d. or rivaroxaban 10 mg o.d. as potential alternatives for patients with CrCl > 30 mL/min who have contraindications or intolerance to LMWH or a previous history of heparin-induced thrombocytopenia. In patients with a CrCl < 30 mL/min, LMWH with dose adjustments (if required) is the preferred prophylaxis* ***(Grade 2B)****.*
      v. *VKAs at doses adapted to obtain a target INR 2.0–3.0 are an option if direct anti-Xas are not available and LMWH are contraindicated.*
8. *We recommend to evaluate the bleeding risk (considering platelet count, hemoglobin levels, renal function, contraindications to anticoagulation, drug–drug interactions, and personal history of clinically relevant hemorrhage) in patients at high/intermediate risk for VTE who are scheduled for pharmacological thromboprophylaxis* ***(Grade 1A)****.*
9. *We recommend the administration of low-dose aspirin for patients classified as low risk* ***(Grade 2B)****.*
10. *We recommend applying pharmacological thromboprophylaxis for at least 6 months from the last risk evaluation. The panel discourages discontinuation of thromboprophylaxis particularly in patients with residual disease receiving treatment with IMiD. Downgrading thromboprophylaxis during maintenance should be considered on a case-by-case basis following risk assessment with the IMPEDE score and careful evaluation of the benefit/risk ratio, patients' preference and impact of the pharmacological thromboprophylaxis on the quality of life of the patient.*
11. *The panel encourages research on the efficacy and safety of thromboprophylaxis with LMWH or direct anti-Xa and the conduction of Phase III studies with risk-stratified patient selection.*
12. *The panel encourages research on the comparison of the existing RAMs and the performance of appropriately designed prospective cohort studies to derive and validate clinico-biological RAMs for accurate identification of MM patients eligible for thromboprophylaxis.*

Thromboprophylaxis in acute medically ill patients with MM

Patients with MM suffering acute medical illness, including COVID-19, benefit equally from the standard thromboprophylaxis practices for acutely ill medical patients and should receive the same thromboprophylaxis as recommended in the most recent guidelines.

Recommendations

We recommend to administer prophylactic doses of LMWH or fondaparinux to patients with active MM and acute medical illness, either hospitalized or receiving care in the primary healthcare setting or at home ***(Grade 1A)****.*

Treatment of VTE in patients with MM

LMWH has been the first choice for the acute phase of thrombosis in patients with malignancies and for long-term secondary prevention for almost 2 decades.[59] Phase III RCTs conducted in patients with CAT showed that the direct anti-Xa agents, apixaban, edoxaban, or rivaroxaban, have similar efficacy and safety compared to LMWHs.[60–63]

The multicenter randomized double-blind, noninferiority trial APICAT compared the efficacy and safety of apixaban at a reduced (2.5 mg) or full (5.0 mg) dose twice daily for 12 months (2.5 mg b.i.d.) in 1766 patients with active cancer and proximal DVT or PE who had completed at least 6 months of anticoagulant.[64] The patients with MM were 3.8% in the reduced-dose group and 4.1% in the full-dose group. The median treatment duration was 11.8 months (interquartile range, 8.3–12.1). Recurrent VTE occurred in 18 patients (cumulative incidence, 2.1%) in the reduced-dose group and in 24 (cumulative incidence, 2.8%) in the full-dose group (adjusted sub-HR, 0.76; 95% CI, 0.41–1.41; P = 0.001 for noninferiority). Clinically relevant bleeding occurred in 102 patients (cumulative incidence, 12.1%) in the reduced-dose group and in 136 (cumulative incidence, 15.6%) in the full-dose group (adjusted sub-HR, 0.75; 95% CI, 0.58–0.97; P = 0.03). Mortality was 17.7% in the reduced-dose group and 19.6% in the full-dose group (adjusted HR, 0.96; 95% CI, 0.86–1.06). Extended anticoagulation with reduced-dose apixaban was noninferior to full-dose apixaban for the prevention of recurrent VTE in patients with active cancer. The reduced dose led to a lower incidence of clinically relevant bleeding complications than the full dose.

Patients with MM were underrepresented in these clinical trials[60–62,65] but overall, VTE management follows the same patterns as in other malignancies.

In selected patients (i.e., elderly and frail patients, patients with renal insufficiency, and patients with liver dysfunction), treatment with LMWH compared to direct anti-Xa agents has the advantage of dose adjustment based on either the body weight or the levels of the

TABLE 4 Treatment options for VTE in cancer patients.

Drug	Thromboprophylaxis	Initial treatment of established VTE (5–10 days)	Long-term phase (first 3–6 months) and extended phase (beyond 6 months)
Heparins (LMWH)			
Dalteparin	5000 anti-Xa IU sc o.d.	100 anti-Xa IU/kg every 12 h, or 200 anti-Xa IU/kg o.d. for the first 30 days	150 anti-Xa IU/kg o.d. after Day 30
Enoxaparin	4000 anti-Xa IU sc o.d.	100 anti-Xa IU/kg every 12 h, or 150 anti-Xa IU/kg o.d.	100 anti-Xa IU/kg every 12 h, or 150 anti-Xa IU/kg o.d.
Tinzaparin	4500 anti-Xa IU sc o.d.	175 anti-Xa IU/kg o.d.	175 anti-Xa IU/kg o.d.
Nadroparin	≤70 kg: 3800 anti-Xa IU sc o.d. >70 kg: 5700 anti-Xa IU sc o.d.	86 anti-Xa IU/kg every 12 h, or 171 anti-Xa IU/kg o.d.	86 anti-Xa IU/kg every 12 h, or 171 anti-Xa IU/kg o.d.
Bemiparin		115 anti-Xa IU/kg o.d.	115 anti-Xa IU/kg o.d.
Heparins (UFH)			
UFH	5000 IU every 8–12 h sc	80 IU/kg iv bolus, then 18 IU/kg/h iv; adjust dose based on aPTT	–
DOACs			
Edoxaban	60 mg o.d. to prevent VTE recurrence	–	60 mg o.d. 30 mg o.d. if: (i) CrCl < 50 mL/min, (ii) ≤60 kg, or (iii) receiving P-gp inhibitors
Rivaroxaban	10 mg o.d.	15 mg every 12 h for 3 weeks	20 mg o.d.
Apixaban	2.5 mg every 12 h	10 mg every 12 h for 7 days	5 mg every 12 h
Vitamin K antagonists			
Acenocoumarol	Adjust dose to maintain INR 2–3	–	Adjust dose to maintain INR 2–3
Phenprocoumon	Adjust dose to maintain INR 2–3	–	Adjust dose to maintain INR 2–3
Warfarin	Adjust dose to maintain INR 2–3	–	Adjust dose to maintain INR 2–3

Note: Agents are presented and categorized with dose presentation for each indication. This table summarizes thromboprophylaxis and treatment dosing strategies for various anticoagulants, including LMWHs, DOACs, UFH, and vitamin K antagonists. It details drug-specific recommendations for prophylaxis, initial treatment (5–10 days), and long-term/extended treatment phases. This resource supports clinicians in selecting and dosing anticoagulation based on clinical phase, renal function, and patient-specific factors.

Abbreviations: aPTT, activated partial thromboplastin time; CrCl, creatinine clearance; DOAC, direct oral anticoagulant; INR, international normalized ratio; iv, intravenous; LMWH, low-molecular-weight heparin; o.d., once daily; UFH, unfractionated heparin; VTE, venous thromboembolism; Xa, activated coagulation factor X.

anti-Xa activity in plasma measured 4 h after the subcutaneous injection. LMWH also presents limited drug-to-drug interactions, particularly relevant when patients receive strong inhibitors or inducers of the cytochrome P450 3A4 (CYP3A4) and P-glycoprotein.

There are no data on the safety of the antithrombotic treatment for secondary prevention of VTE in MM patients with renal impairment. Post hoc and subgroup analyses suggest that LMWH and direct factor Xa inhibitors in patients with moderate renal impairment (CrCl 30–60 mL/min) can be administered without dose modifications as the safety and efficacy profile does not change. Patients with severe renal impairment (CrCl < 30 mL/min) were excluded from the randomized Phase III clinical trials, and data for anti-Xa agents in this setting are lacking. Therefore, either unfractionated heparin (UFH) followed by VKA or LMWH with the dose adjusted to anti-Xa activity level should be applied. For patients with severe thrombocytopenia ($<50 \times 10^9$/L), if there is a high risk of thrombosis progression, full-dose anticoagulation with platelet transfusion support can be administered, or for lower risk cases, intermediate or prophylactic dose LMWH is an option.

The optimal management of superficial vein thrombosis has not been established yet, and data in MM patients are lacking. Superficial vein thrombosis (SVT) of the long saphenous vein within 3 cm of the saphenofemoral junction (SFJ) is considered equivalent to a DVT. For less severe forms of SVT, the recommendation would be therapeutic anticoagulation for 6 weeks at least and then switch to prophylaxis, but there are no MM-specific data to guide clinical practice.

Recommendations

i. *Treatment of the acute phase of VTE in patients with MM.*
   1. *We recommend a systematic evaluation of the bleeding risk before the initiation of antithrombotic therapy. Predictors of bleeding risk are: renal insufficiency, low platelet count (<50 G/L), anemia, and drug–drug interactions (for the direct anti-Xa agents) (**Grade 1A**).*
   2. *We recommend, without interruption of the antimyeloma therapy,*
      a. *the initiation of a therapeutic dose of LMWH in patients at high bleeding risk (Grade 1A);*
      b. *the initiation of therapeutic doses of apixaban, edoxaban, or rivaroxaban in patients with low bleeding risk and low potential of drug–drug interactions (edoxaban requires lead-in therapy with LMWH for the first 5–10 days) (Grade 1A);*
      c. *the initiation of therapeutic doses of UFH with regular monitoring of the anti-Xa levels in plasma in patients with severe renal insufficiency (CrCl < 30 mL/min) (**Grade 1A**).*
   3. *We suggest, in the case of life-threatening or recurrent VTE during IMiD/carfilzomib-based regimens, individualized therapeutic decisions. Factors to be considered in individualized therapeutic strategy include the depth of response to the current regimen, the severity of the thrombotic event, and the patient's risk profile recurrent VTE (**Grade 2C**).*

ii. *Secondary prevention of VTE in patients with MM.*
   1. We recommend for secondary prevention of VTE treatment with LMWH or anti-Xa agents (apixaban, edoxaban, or rivaroxaban) **(Grade 1A)**.
      a. The optimal antithrombotic agent should be selected based on the renal function and the potential drug–drug interactions **(Grade 1B)**.
   2. We recommend a minimum duration of 6 months for antithrombotic treatment in patients with VTE and MM *(Grade 1A)*.
      - Active MM is defined as (i) NDMM, (ii) MM at relapse/progression, and (iii) MM disease that is not in remission despite treatment administration.
   3. We recommend extension of the antithrombotic treatment beyond 6 months in patients with active MM (Grade 1A).
   4. We recommend the continuation of the antithrombotic treatment beyond 6 months in patients receiving ongoing IMiD or carfilzomib therapy.
   5. We recommend for patients with active MM that the decision for the extension of antithrombotic treatment beyond 6 months should involve careful evaluation of VTE recurrence risk versus bleeding risk *(Grade 1A)*.
      a. We suggest referring the patient to a thrombosis and hemostasis specialist. Periodic reevaluation of thrombotic and hemorrhagic risk is necessary during long-term antithrombotic treatment ***(Grade 2B)***.
   6. *We* recommend reduced-dose apixaban (2.5 mg twice daily) over full-dose apixaban (5 mg twice daily) for MM patients requiring extended secondary prevention of CAT beyond 6 months post-index event *(Grade 1B)*.

Prophylaxis and treatment of catheter-related VTE

Evidence is scarce among cancer patients, and there are no data specific for MM. There is one prospective trial that evaluated LMWH plus VKA in cancer patients with catheter-related VTE. There is no robust data yet concerning the role of factor Xa inhibitors.

Recommendations

1. *Routine pharmacological prophylaxis is not recommended for the prevention of catheter-related VTE (**Grade 2C**).*
2. *For symptomatic catheter-related VTE in patients with MM, anticoagulation is recommended for a minimum of 3 months (Grade 2a). The preferred first-line therapy is LMWH, but anti-Xa agents or VKAs are alternative options (**Grade 2C**).*
3. *We recommend antithrombotic treatment for a minimum of 3 months for symptomatic catheter-related VTE in patients with MM, anticoagulation (*Grade 2A*). The preferred first-line therapy is LMWH, with direct oral anti-Xa agents or VKAs as alternative options (*Grade 2C*).*
4. *We suggest evaluation of the risk of myeloma-associated VTE after 3 months of the antithrombotic treatment and application of the recommended thromboprophylaxis if the patient is stratified at high/intermediate risk (**Grade 2C**).*
5. *Overall, further MM-specific recommendations cannot be extrapolated. The panel recommends that these patients are managed according to standard clinical practice for non-MM patients and encourages the design of trials that could provide data regarding CVC use in MM patients.*

DISCUSSION

The EMN-VTE Consensus Guidelines mark a significant advancement in VTE management for MM patients. These recommendations were developed through rigorous multidisciplinary collaboration of experts in thrombosis and specialists in MM, under the auspices of the EMN. This expert consensus provides tailored, evidence-based strategies that address the unique thrombotic risks inherent to myeloma pathophysiology and its treatments.

This document introduces key updates, including the simplification of risk stratification by grouping intermediate- and high-risk patients, and advocates for the use of factor Xa inhibitors alongside LWMH for primary thromboprophylaxis. The panel recommends universal thromboprophylaxis for all newly diagnosed or relapsed patients initiating treatment with IMiDs or carfilzomib combinations, regardless of baseline risk scores, following a careful hemorrhagic risk evaluation. This recommendation is supported by data showing persistently high VTE rates in patients receiving IMiDs, even when stratified by risk models, and recognizes that current risk assessment

tools were developed before the use of carfilzomib. By integrating disease activity and contemporary treatment regimens into its framework, the EMN-VTE guideline addresses unmet clinical needs and aims to harmonize and improve real-world thromboprophylaxis practices.

The key points addressed by the Expert Panel Consensus Guidance are as follows:

1. VTE is a frequent and life-threatening complication of MM that negatively impacts patients' quality of life, interferes with the scheduled treatment, and decreases life expectancy.
2. Hematologists, angiologists, thrombosis, and hemostasis experts and general practitioners should do the following:
   - Inform patients about the risk of VTE, educate them about the clinical signs and symptoms, target modifiable risk factors, and inform them about the pathways for prompt diagnosis and treatment.
   - Perform routine and regular evaluation of VTE risk by applying recommended RAMs, acknowledging that the accuracy of these scores is suboptimal.
   - Evaluate regularly the bleeding risk, acknowledging that a validated RAM for bleeding risk in ambulatory cancer patients eligible for pharmacological thromboprophylaxis is not yet available.
   - Initiate, for intermediate and high-risk patients, pharmacological thromboprophylaxis with LMWH or factor Xa inhibitors (apixaban or rivaroxaban) at the recommended prophylactic doses.
   - Initiate low-dose ASA in patients classified as low risk.
   - Initiate pharmacological thromboprophylaxis with LMWH or factor Xa inhibitors (apixaban or rivaroxaban) at the recommended prophylactic doses for all patients due to start IMiD-based or carfilzomib-based therapy.
   - Pharmacological thromboprophylaxis should be administered for at least 6 months, followed by a case-by-case evaluation.
   - Predefine a local healthcare pathway—from the primary healthcare structures up to specialized academic centers—for prompt diagnosis and treatment of VTE in ambulatory patients with MM.
   - Initiate antithrombotic treatment with therapeutic doses of LMWH or factor Xa inhibitors in patients with diagnosed VTE for at least 6 months.
   - Privilege low-dose apixaban 2.5 mg twice daily over full-dose apixaban (5 mg twice daily) for patients eligible for long-term secondary prevention of VTE after 6 months from the index event.
   - Apply a personalized strategy for the long-term anticoagulation for secondary prevention of VTE taking into consideration the disease status, hematological response, and bleeding risks.
   - The choice of the optimal antithrombotic agent, either for primary thromboprophylaxis or VTE treatment, should take into consideration patients' preference, impact on quality of life, and drug-to-drug interactions.

Given the complexity of managing patients with MM, whether at risk of VTE or already diagnosed with VTE, along with the dynamic nature of disease progression, evolving treatment regimens, and the aging population, the panel strongly recommends establishing specialized outpatient Thrombosis Consultations. We require more effective and updated VTE risk assessment tools, developed prospectively based on cohorts that include triplet/quadruplet novel agent combinations to reflect current clinical practice accurately. In addition, prospective randomized trials are required to provide answers to the question of optimal thromboprophylaxis modality and duration, based on MM-specific patient profiles. VTE is likely to remain an ongoing risk for MM patients, and as a community, we need to implement consistent clinical practices to improve outcomes.

AUTHOR CONTRIBUTIONS

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CONFLICT OF INTEREST STATEMENT

Grigoris Gerotziafas: no relevant conflicts. Despina Fotiou: speaker honoraria from Sanofi, Janssen. Inger Nijhof: advisory board or speaker honoraria funding from Janssen Pharmaceuticals, Takeda, BMS, and Kite Pharma; all paid to institution. Cihan Ay: personal fees for lectures and/or participation in advisory boards from Bayer, BMS/Pfizer, Daiichi-Sankyo, and Sanofi. Ramon Lecumberri: no relevant conflicts. Alessandra Laroca: advisory board for Janssen, GSK, Pfizer, Sanofi, Takeda, Menarini, and Oncopeptides; honoraria from Amgen, Janssen, GSK, Pfizer, Sanofi, Takeda, and Menarini. Gordon Cook: research funding from J&J, Takeda, and BMS; consultancy for J&J, Pfizer, AZ, and Menarini Stemline; speaker bureau for J&J, Pfizer, and Menarini/Stemline. Monika Engelhardt: advisory for Amgen, Bristol Myers Squibb, Janssen, GSK, Oncopeptides, Pfizer, Sanofi, and Stemline; honoraria from Amgen, AbbVie, Bristol Myers Squibb,

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Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

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REFERENCES

1. Cowan AJ, Green DJ, Kwok M, et al. Diagnosis and management of multiple myeloma: a review. *JAMA*. 2022;327(5):464-477. doi:10.1001/jama.2022.0003
2. Brown JD, Adams VR. Incidence and risk factors of thromboembolism with multiple myeloma in the presence of death as a competing risk: an empirical comparison of statistical methodologies. *Healthcare (Basel)*. 2016;4(1):16. doi:10.3390/healthcare4010016
3. Piedra K, Peterson T, Tan C, et al. Comparison of venous thromboembolism incidence in newly diagnosed multiple myeloma patients receiving bortezomib, lenalidomide, dexamethasone (RVD) or carfilzomib, lenalidomide, dexamethasone (KRD) with aspirin or rivaroxaban thromboprophylaxis. *Br J Haematol*. 2022;196(1):105-109. doi:10.1111/bjh.17772
4. Charalampous C, Goel U, Kapoor P, et al. Thrombosis in multiple myeloma: risk estimation by induction regimen and association with overall survival. *Am J Hematol*. 2023;98(3):413-420. doi:10.1002/ajh.26806
5. Leleu X, Rodon P, Hulin C, et al. MELISSE, a large multicentric observational study to determine risk factors of venous thromboembolism in patients with multiple myeloma treated with immunomodulatory drugs. *Thromb Haemost*. 2013;110(4):844-851. doi:10.1160/TH13-02-0140
6. Kristinsson SY. Thrombosis in multiple myeloma. *Hematology*. 2010;2010:437-444. doi:10.1182/asheducation-2010.1.437
7. Za T, De Stefano V, Rossi E, et al. Arterial and venous thrombosis in patients with monoclonal gammopathy of undetermined significance: incidence and risk factors in a cohort of 1491 patients. *Br J Haematol*. 2013;160(5):673-679. doi:10.1111/bjh.12168
8. Rögnvaldsson S, Gasparini A, Thorsteinsdottir S, et al. Monoclonal gammopathy of undetermined significance and the risk of thrombotic events: results from istopmm, a population-based screening study in Iceland. *Blood*. 2023;142(suppl 1):216.
9. Kristinsson SY, Pfeiffer RM, Bjorkholm M, Schulman S, Landgren O. Thrombosis is associated with inferior survival in multiple myeloma. *Haematologica*. 2012;97(10):1603-1607. doi:10.3324/haematol.2012.064444
10. Schoen MW, Carson KR, Luo S, et al. Venous thromboembolism in multiple myeloma is associated with increased mortality. *Res Pract Thromb Haemost*. 2020;4(7):1203-1210. doi:10.1002/rth2.12411
11. Khorana A, Dalal MR, Lin J, Connolly GC. Health care costs associated with venous thromboembolism in selected high-risk ambulatory patients with solid tumors undergoing chemotherapy in the United States. *Clinicoecon Outcomes Res*. 2013;5:101-108. doi:10.2147/CEOR.S39964
12. Rajkumar SV. Value and cost of myeloma therapy. *Am Soc Clin Oncol Educ Book*. 2018;38:662-666. doi:10.1200/EDBK_200867
13. Schünemann HJ, Wiercioch W, Brozek J, et al. GRADE Evidence to Decision (EtD) frameworks for adoption, adaptation, and de novo development of trustworthy recommendations: GRADE-ADOLOPMENT. *J Clin Epidemiol*. 2017;81:101-110. doi:10.1016/j.jclinepi.2016.09.009
14. Fotiou D, Sergentanis TN, Papageorgiou L, et al. Longer procoagulant phospholipid-dependent clotting time, lower endogenous thrombin potential and higher tissue factor pathway inhibitor concentrations are associated with increased VTE occurrence in patients with newly diagnosed multiple myeloma: results of the prospective ROADMAP-MM-CAT study. *Blood Cancer J*. 2018;8(11):102. doi:10.1038/s41408-018-0135-y
15. Huang H, Li H, Li D. Effect of serum monoclonal protein concentration on haemostasis in patients with multiple myeloma. *Blood Coagul Fibrinolysis*. 2015;26(5):555-559. doi:10.1097/MBC.0000000000000296

16. Brown JD, Adams VR, Moga DC. Impact of time-varying treatment exposures on the risk of venous thromboembolism in multiple myeloma. *Healthcare (Basel)*. 2016;4(4):93. doi:10.3390/healthcare4040093
17. Zonder JA, Crowley J, Hussein MA, et al. Lenalidomide and high-dose dexamethasone compared with dexamethasone as initial therapy for multiple myeloma: a randomized Southwest Oncology Group trial (S0232). *Blood*. 2010;116(26):5838-5841. doi:10.1182/blood-2010-08-303487
18. Palumbo A, Palladino C. Venous and arterial thrombotic risks with thalidomide: evidence and practical guidance. *Ther Adv Drug Saf*. 2012;3(5):255-266. doi:10.1177/2042098612452291
19. Rajkumar SV, Jacobus S, Callander NS, et al. Lenalidomide plus high-dose dexamethasone versus lenalidomide plus low-dose dexamethasone as initial therapy for newly diagnosed multiple myeloma: an open-label randomised controlled trial. *Lancet Oncol*. 2010;11(1):29-37. doi:10.1016/S1470-2045(09)70284-0
20. Palumbo A, Hajek R, Delforge M, et al. Continuous lenalidomide treatment for newly diagnosed multiple myeloma. *N Engl J Med*. 2012;366(19):1759-1769. doi:10.1056/NEJMoa1112704
21. Scott LJ. Pomalidomide: a review of its use in patients with recurrent multiple myeloma. *Drugs*. 2014;74(5):549-562. doi:10.1007/s40265-014-0196-6
22. Zangari M, Fink L, Zhan F, Tricot G. Low venous thromboembolic risk with bortezomib in multiple myeloma and potential protective effect with thalidomide/lenalidomide-based therapy: review of data from phase 3 trials and studies of novel combination regimens. *Clin Lymphoma Myeloma Leuk*. 2011;11(2):228-236. doi:10.1016/j.clml.2011.03.006
23. Kumar SK, Jacobus SJ, Cohen AD, et al. Carfilzomib or bortezomib in combination with lenalidomide and dexamethasone for patients with newly diagnosed multiple myeloma without intention for immediate autologous stem-cell transplantation (ENDURANCE): a multicentre, open-label, phase 3, randomised, controlled trial. *Lancet Oncol*. 2020;21(10):1317-1330. doi:10.1016/S1470-2045(20)30452-6
24. Costa BA, Costa TA, Saravia SD, et al. Thromboembolic risk of carfilzomib or bortezomib in combination with lenalidomide and dexamethasone for newly diagnosed multiple myeloma: a comparative systematic review and meta-analysis. *Am J Hematol*. 2024;99(6):1056-1065. doi:10.1002/ajh.27288
25. Loncharich AJ, Fiala MA, Slade MJ, et al. Venous thromboembolism risk in patients with newly diagnosed multiple myeloma treated with carfilzomib or bortezomib in combination with lenalidomide and dexamethasone. *Clin Lymphoma Myeloma Leuk*. 2023;23(11):825-828. doi:10.1016/j.clml.2023.07.009
26. Camilleri M, Cuadrado M, Phillips E, et al. Thrombotic microangiopathy in untreated myeloma patients receiving carfilzomib, cyclophosphamide and dexamethasone on the CARDAMON study. *Br J Haematol*. 2021;193(4):750-760. doi:10.1111/bjh.17377
27. Gay F, Günther A, Offidani M, et al. Carfilzomib, bendamustine, and dexamethasone in patients with advanced multiple myeloma: the EMN09 phase 1/2 study of the European Myeloma Network. *Cancer*. 2021;127(18):3413-3421. doi:10.1002/cncr.33647
28. Wang J, Park C, Arroyo-Suarez R. Venous thromboembolism in patients with multiple myeloma receiving daratumumab-based regimens: a post hoc analysis of phase 3 clinical trials. *Leuk Lymphoma*. 2021;62(9):2219-2226. doi:10.1080/10428194.2021.1910687
29. Sborov D, Baljevic M, Reeves B, et al. Daratumumab (DARA) plus lenalidomide, bortezomib, and dexamethasone (RVd) in newly diagnosed multiple myeloma (NDMM): analysis of vascular thrombotic events (VTEs) in the GRIFFIN study [abstract]. *Clin Lymphoma Myeloma*. 2021;21(2):135-136.
30. Chakraborty R, Bin Riaz I, Malik SU, et al. Venous thromboembolism risk with contemporary lenalidomide-based regimens despite thromboprophylaxis in multiple myeloma: a systematic review and meta-analysis. *Cancer*. 2020;126(8):1640-1650. doi:10.1002/cncr.32682
31. Bindal P, Patell R, Chiasakul T, et al. A meta-analysis to assess the risk of bleeding and thrombosis following chimeric antigen receptor T-cell therapy: communication from the ISTH SSC Subcommittee on Hemostasis and Malignancy. *J Thromb Haemost*. 2024;22(7):2071-2080. doi:10.1016/j.jtha.2024.03.021
32. Cini M, Zamagni E, Valdré L, et al. Thalidomide-dexamethasone as up-front therapy for patients with newly diagnosed multiple myeloma: thrombophilic alterations, thrombotic complications, and thromboprophylaxis with low-dose warfarin. *Eur J Haematol*. 2010;84(6):484-492. doi:10.1111/j.1600-0609.2010.01434.x
33. Nicolaides AN, Fareed J, Spyropoulos AC, et al. Prevention and management of venous thromboembolism. International Consensus Statement. Guidelines according to scientific evidence. *Int Angiol*. 2024;43(1):1-222. doi:10.23736/S0392-9590.23.05177-5
34. Falanga A, Marchetti M. Venous thromboembolism in the hematologic malignancies. *J Clin Oncol*. 2009;27(29):4848-4857. doi:10.1200/JCO.2009.22.8197
35. Palumbo A, Rajkumar SV, San Miguel JF, et al. International Myeloma Working Group consensus statement for the management, treatment, and supportive care of patients with myeloma not eligible for standard autologous stem-cell transplantation. *J Clin Oncol*. 2014;32(6):587-600. doi:10.1200/JCO.2013.48.7934
36. Terpos E, Kleber M, Engelhardt M, et al. European Myeloma Network guidelines for the management of multiple myeloma-related complications. *Haematologica*. 2015;100(10):1254-1266. doi:10.3324/haematol.2014.117176
37. Callander NS, Baljevic M, Adekola K, et al. NCCN Guidelines® insights: multiple myeloma, version 3.2022: featured updates to the NCCN guidelines. *J Natl Compr Canc Netw*. 2022;20(1):8-19. doi:10.6004/jnccn.2022.0002
38. Bradbury CA, Craig Z, Cook G, et al. Thrombosis in patients with myeloma treated in the Myeloma IX and Myeloma XI phase 3 randomized controlled trials. *Blood*. 2020;136(9):1091-1104. doi:10.1182/blood.2020005125
39. Sanfilipo K, Carson K, BF C. Aspirin may be inadequate thromboprophylaxis in multiple myeloma. *Blood*. 2017;130(3):3419.
40. Li A, Wu Q, Luo S, et al. Derivation and validation of a risk assessment model for immunomodulatory drug-associated thrombosis among patients with multiple myeloma. *J Natl Compr Canc Netw*. 2019;17(7):840-847. doi:10.6004/jnccn.2018.7273
41. Chakraborty R, Rybicki L, Wei W, et al. Abnormal metaphase cytogenetics predicts venous thromboembolism in myeloma: derivation and validation of the PRISM score. *Blood*. 2022;140:2443-2450.
42. Zweegman S, van de Donk NWCJ. Supportive care in myeloma-when treating the clone alone is not enough. *Hematology*. 2024;2024(1):569-581. doi:10.1182/hematology.2024000579
43. Bradbury C. Thrombotic events in patients with myeloma treated with immunomodulatory drugs; results of the myeloma XI study. *Blood*. 2017;130:553.
44. Paludo J, Mikhael JR, LaPlant BR, et al. Pomalidomide, bortezomib, and dexamethasone for patients with relapsed lenalidomide-refractory multiple myeloma. *Blood*. 2017;130(10):1198-1204. doi:10.1182/blood-2017-05-782961
45. Kahale LA, Matar CF, Tsolakian I, et al. Antithrombotic therapy for ambulatory patients with multiple myeloma receiving immunomodulatory agents. *Cochrane Database Syst Rev*. 2021;9:014739. doi:10.1002/14651858.CD014739
46. Larocca A, Cavallo F, Bringhen S, et al. Aspirin or enoxaparin thromboprophylaxis for patients with newly diagnosed multiple myeloma treated with lenalidomide. *Blood*. 2012;119(4):933-939; quiz 1093. doi:10.1182/blood-2011-03-344333
47. Palumbo A, Cavo M, Bringhen S, et al. Aspirin, warfarin, or enoxaparin thromboprophylaxis in patients with multiple myeloma treated with thalidomide: a phase III, open-label, randomized trial. *J Clin Oncol*. 2011;29(8):986-993. doi:10.1200/JCO.2010.31.6844

48. Sayar Z, Gates C, Bristogiannis S, et al. Safety and efficacy of apixaban as thromboprophylaxis in myeloma patients receiving chemotherapy: a prospective cohort study. *Thromb Res*. 2022;213:27-29. doi:10.1016/j.thromres.2022.03.001
49. Costa TA, Felix N, Costa BA, Godoi A, Nogueira A, Rossi A. Direct oral anticoagulants versus aspirin for primary thromboprophylaxis in patients with multiple myeloma undergoing outpatient therapy: a systematic review and updated meta-analysis. *Br J Haematol*. 2023;203(3):395-403. doi:10.1111/bjh.19017
50. Zoppellaro G, Veronese N, Granziera S, Gobbi L, Stubbs B, Cohen AT. Primary thromboembolic prevention in multiple myeloma patients: an exploratory meta-analysis on aspirin use. *Sem Hematol*. 2018;55(4):182-184. doi:10.1053/j.seminhematol.2017.08.002
51. Carrier M, Abou-Nassar K, Mallick R, et al Apixaban to prevent venous thromboembolism in patients with cancer. *N Engl J Med*. 2018;380:711-719. doi:10.1056/NEJMoa1814468
52. Khorana AA, Soff GA, Kakkar AK, et al. Rivaroxaban for thromboprophylaxis in high-risk ambulatory patients with cancer. *N Engl J Med*. 2019;380(8):720-728. doi:10.1056/NEJMoa1814630
53. Pegourie B, Karlin L, Benboubker L, et al. Apixaban for the prevention of thromboembolism in immunomodulatory-treated myeloma patients: myelaxat, a phase 2 pilot study. *Am J Hematol*. 2019;94(6):635-640. doi:10.1002/ajh.25459
54. Cornell RF, Goldhaber SZ, Engelhardt BG, et al. Primary prevention of venous thromboembolism with apixaban for multiple myeloma patients receiving immunomodulatory agents. *Br J Haematol*. 2020;190(4):555-561. doi:10.1111/bjh.16653
55. Storrar NPF, Mathur A, Johnson PRE, Roddie PH. Safety and efficacy of apixaban for routine thromboprophylaxis in myeloma patients treated with thalidomide- and lenalidomide-containing regimens. *Br J Haematol*. 2019;185(1):142-144. doi:10.1111/bjh.15392
56. Man L, Morris A, Brown J, Palkimas S, Davidson K. Use of direct oral anticoagulants in patients on immunomodulatory agents. *J Thromb Thrombolysis*. 2017;44(3):298-302. doi:10.1007/s11239-017-1534-9
57. Blimark C, Holmberg E, Mellqvist UH, et al. Multiple myeloma and infections: a population-based study on 9253 multiple myeloma patients. *Haematologica*. 2015;100(1):107-113. doi:10.3324/haematol.2014.107714
58. Sborov DW, Baljevic M, Reeves B, et al. Daratumumab plus lenalidomide, bortezomib and dexamethasone in newly diagnosed multiple myeloma: analysis of vascular thrombotic events in the GRIFFIN study. *Br J Haematol*. 2022;199(3):355-365. doi:10.1111/bjh.18432
59. Kearon C, Akl EA, Ornelas J, et al. Antithrombotic therapy for VTE disease. *Chest*. 2016;149(2):315-352. doi:10.1016/j.chest.2015.11.026
60. McBane RD, Wysokinski WE, Le-Rademacher JG, et al. Apixaban and dalteparin in active malignancy-associated venous thromboembolism: the ADAM VTE trial. *J Thromb Haemostasis*. 2020;18(2):411-421. doi:10.1111/jth.14662
61. Agnelli G, Becattini C, Meyer G, et al. Apixaban for the treatment of venous thromboembolism associated with cancer. *N Engl J Med*. 2020;382(17):1599-1607. doi:10.1056/NEJMoa1915103
62. Cohen AT, Dhamane AD, Liu X, et al. Effectiveness and safety of extended treatment apixaban versus low-molecular-weight heparin in cancer-associated venous thromboembolism. *J Natl Compr Canc Netw*. 2024;22(6):397-403. doi:10.6004/jnccn.2024.7016
63. Raskob GE, van Es N, Verhamme P, et al. Edoxaban for the treatment of cancer-associated venous thromboembolism. *N Engl J Med*. 2018;378(7):615-624. doi:10.1056/NEJMoa1711948
64. Mahé I, Carrier M, Mayeur D, et al. Extended reduced-dose apixaban for cancer-associated venous thromboembolism. *N Engl J Med*. 2025;392(14):1363-1373. doi:10.1056/NEJMoa2416112
65. Young AM, Marshall A, Thirlwall J, et al. Comparison of an oral factor Xa inhibitor with low molecular weight heparin in patients with cancer with venous thromboembolism: results of a randomized trial (SELECT-D). *J Clin Oncol*. 2018;36(20):2017-2023. doi:10.1200/JCO.2018.78.8034